

CMB Polarization

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The First Light of Universe: An Introduction to Cosmic Microwave Background

In 1964, while working at Bell Laboratories A. Penzias and R. Wilson faced an unexpected issue. Their horn antenna consistently recorded an excess noise temperature of about 3 K. They could not eliminate this signal, nor could they find any known source for it. Most confusingly, this excess noise was isotropic; regardless of where they pointed the antenna, the signal remained the same. This unexplained temperature excess became one of the most significant discoveries in modern cosmology: the Cosmic Microwave Background (CMB). It is the oldest light in the Universe, released when radiation could finally travel freely without frequent interactions with matter, around 380,000 years after the Big Bang.

In this article, we will explore the origin of the CMB, its key observed properties with a focus on polarization, which is an active area of research in modern cosmology. To understand these ideas, we first need to start with a discussion on basic cosmology. The sources primarily used in this article are [2, 7, 3, 5, 6, 8, 10, 11]

The Standard Cosmological Paradigm

As the opening lines of the famous sitcom *The Big Bang Theory* put it:
*Our whole Universe was in a hot dense state,
Then nearly fourteen billion years ago expansion started...*

Indeed, the modern description of our Universe begins with the Big Bang model, which postulates that space itself has been expanding from an initially hot and dense state. This expansion is encoded in the scale factor ‘ a ’, which describes how physical distances grow with cosmic time. The dynamics of $a(t)$ are governed by the Friedmann equations, derived from Einstein’s equations of General

Relativity under the assumption of large-scale homogeneity and isotropy. While we will not dwell on their details here, their key implication is clear: the Universe is expanding, and its contents cool as it does so.

Inflation and the Early Universe

According to the standard cosmological picture, the early Universe underwent a phase of extremely rapid expansion known as cosmic inflation, occurring roughly 10^{-36} to 10^{-32} seconds after the Big Bang. During this brief interval the scale factor increased e^{60} to e^{70} times. As a result, regions that were initially microscopically small were stretched to astronomical sizes. While the precise physical origin of inflation is still not fully understood, this phase plays a central role in shaping the large-scale structure of the Universe and will become important later in our discussion.

Following inflation, the standard model particles formed and the Universe became hot and dense. At these early times, the temperature was so high that atoms could not exist. All baryonic matter was in the form of a fully ionized plasma consisting of protons, electrons, and other light nuclei, immersed in an enormous bath of photons. The photons vastly outnumbered the baryons with a ratio of a billion to one.

As the matter in the early Universe was completely ionized, there existed an abundance of free electrons. These free electrons scattered photons via Thomson scattering, causing photon–electron interactions to occur extremely frequently. As a result, radiation could not propagate freely through space. Instead, photons and baryonic matter formed a tightly coupled plasma, continuously exchanging energy. This caused matter and radiation to be in thermal equilibrium throughout this early epoch.

Recombination and the Release of CMB

As the Universe expanded, it continued to cool. Eventually, at a temperature of about 3000 K, corresponding to a cosmic age of approximately 380,000 years, electrons and protons were able to

combine to form neutral hydrogen atoms. This transition period is known as the Recombination. As recombination progressed and free electrons rapidly disappeared, the Thomson scattering rate dropped dramatically. Photons were no longer frequently scattered, and their mean free path increased rapidly. Radiation effectively decoupled from matter and began to travel freely through space. It is these photons which now make up the Cosmic Microwave Background.

The region in spacetime from which these photons originate is often referred to as the surface of last scattering, though it is in fact a finite thickness in time rather than a sharp surface. Because these photons last scattered when the plasma was in near-perfect thermal equilibrium, the CMB possesses an almost ideal blackbody spectrum.

Since their release, the Universe has expanded by a factor of about ~ 1100 , redshifting the CMB photons from visible/infrared energies down to the microwave regime.

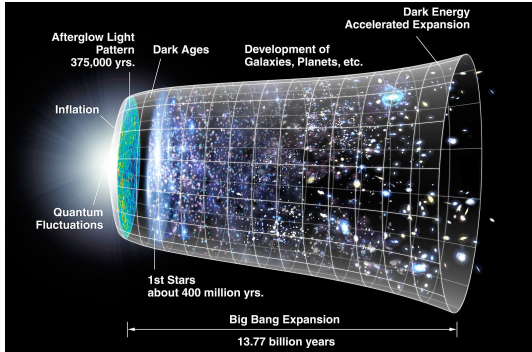


Figure 1: Timeline of the expansion of the universe (Image credits: [Wikipedia](#))

Observational Confirmations

Satellite missions such as COBE, WMAP, and Planck have measured the CMB spectrum extraordinary precision. The observed spectrum is an almost perfect blackbody with a present-day temperature of $T_0 = 2.725 \text{ K}$. This is precisely the excess temperature detected by Penzias and Wilson in 1964. It should now be evident why this radiation is observed from all directions: it was released everywhere throughout the Universe at the 'last scattering', and we are immersed within it.

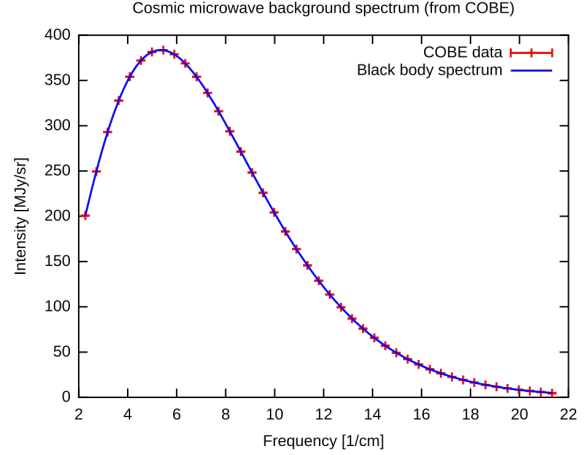


Figure 2: The COBE Spectral data showing the nearly perfect blackbody fit (Image credits: [Wikipedia](#))

Temperature Anisotropies and the Angular Power Spectrum

So far, we have described the Cosmic Microwave Background as a nearly homogeneous bath of radiation with a blackbody spectrum at a temperature of 2.7 K. However, precise observations reveal that the temperature of the CMB is not exactly the same in all directions. Instead, it exhibits small spatial variations across the sky, known as *temperature anisotropies*. The relative magnitude of these fluctuations is

$$\frac{\delta T}{T} \sim 10^{-5}, \quad (1)$$

indicating that the deviations from homogeneity are extremely small, yet non-zero.

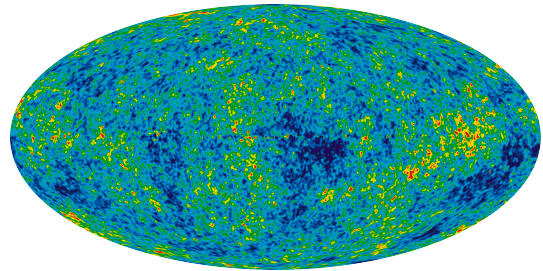


Figure 3: Full-sky map of CMB temperature anisotropies with the mean temperature subtracted. The colour scale shows temperature fluctuations δT at the level of tens of microkelvin. (Image credits: [Wikipedia](#))

These inhomogeneities are believed to have originated during cosmic inflation. During this period, quantum fluctuations of the inflation field

were stretched to macroscopic scales. These fluctuations later manifested as small density and velocity perturbations in the photon-baryon plasma. At recombination, when photons decoupled from matter, these perturbations were imprinted onto the radiation field. We now observe these perturbations as temperature anisotropy in the CMB.

Angular Structure of the Anisotropies

The temperature anisotropies are distributed across a wide range of angular scales on the sky. When we examine the CMB at very large angular scales, we observe broad, sweeping temperature variations. As we probe progressively smaller patches of the sky, finer structures begin to emerge. This scale-dependent behavior tells us that we need a framework that can systematically capture the distribution of fluctuations across all angular scales.

To describe this structure quantitatively, it is natural to decompose the temperature field into spherical harmonics,

$$\frac{\delta T(\hat{n})}{T} = \sum_{\ell, m} a_{\ell m} Y_{\ell m}(\hat{n}), \quad (2)$$

where $\hat{n}$ denotes a direction on the sky. This is analogous to decomposing a periodic function into Fourier components. The coefficients $a_{\ell m}$ quantify the contribution of each harmonic mode to the observed temperature pattern. Each multipole moment ℓ corresponds approximately to angular variations on a scale

$$\theta \sim \frac{\pi}{\ell}. \quad (3)$$

Thus, low values of ℓ describe temperature fluctuations extending over large angular regions, while higher ℓ probe progressively smaller angular scales.

The Angular Power Spectrum

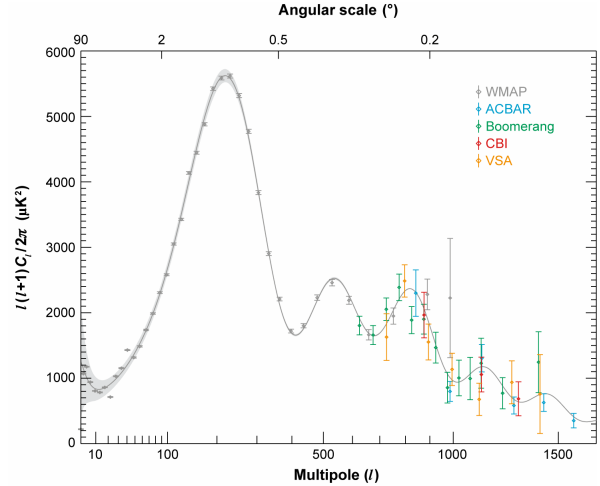
Rather than cataloging every hot and cold spot on the CMB sky, we characterize their distribution statistically through the angular power spectrum. Think of this as measuring how much temperature variation exists at each angular scale, from the largest structures spanning the entire sky down to the smallest details we can resolve.

To put this in mathematical terms, we define the angular power spectrum as

$$C_\ell = \langle |a_{\ell m}|^2 \rangle, \quad (4)$$

where the angular brackets denote an average over all values of m for a given multipole ℓ .

The quantity C_ℓ measures how much power (or variance) the temperature fluctuations carry at angular scale ℓ . This gives us a complete statistical description of the CMB anisotropies without needing to track individual hot and cold spots.



Samtleben D, et al. 2007.
Annu. Rev. Nucl. Part. Sci. 57:245–83

Figure 4: Observed angular power spectrum of CMB temperature anisotropies, plotted as $\ell(\ell + 1)C_\ell/2\pi$ versus the multipole moment ℓ . The factor $\ell(\ell + 1)/2\pi$ approximately flattens the spectrum at large ℓ , making the oscillatory structure more apparent. (Image credits [9])

When we plot the observed power spectrum, something remarkable emerges. Instead of a smooth curve, we see a distinctive pattern of peaks and valleys. This is not measurement error or instrumental noise. These oscillations are real physical features that encode fundamental information about conditions in the early Universe.

Acoustic Oscillations

What causes this oscillatory pattern? The answer lies in sound waves (longitudinal pressure waves) that propagated through the primordial plasma before recombination.

Recall that before recombination, photons and baryons were tightly coupled through Thomson scattering, forming a single fluid. Any small density perturbation in this fluid led to two competing interactions. Gravity acted to pull more matter into the overdense region, trying to enhance the perturbation. But the immense radiation pressure from the trapped photons pushed back, resisting gravitational collapse.

This competition did not lead to equilibrium. Instead, it set the plasma into oscillation. Overdense regions compressed under gravity, building up pressure until the photon force became strong enough to reverse the collapse. The plasma then expanded, overshooting equilibrium, which allowed gravity to pull it back again. These oscillations were sound waves rippling through the photon-baryon fluid.

Different perturbation modes oscillated at differ-

ent frequencies, determined by their wavelength. Consider a perturbation of a particular physical size. From the moment it was created (say, during inflation) until recombination, it had a fixed amount of time to oscillate. Depending on its wavelength, it might complete half an oscillation, a full oscillation, one and a half oscillations, and so on.

At recombination, photons decoupled from matter. Thus the oscillations in baryon-photon plasma were instantly frozen into the CMB radiation. Perturbations that happened to be at an extremum of their oscillation cycle at that moment left the strongest imprint. Modes at maximum compression created the hottest spots, while modes at maximum rarefaction created the coldest spots. These extrema show up as peaks in the angular power spectrum.

The first acoustic peak at $\ell \approx 220$ corresponds to perturbations that completed exactly half an oscillation by recombination, reaching maximum compression just as the photons decoupled. The second peak comes from modes that completed a full oscillation and were maximally compressed again. The third peak corresponds to one and a half oscillations, and so on. The valleys between peaks represent modes caught at intermediate phases, contributing less power.

Polarization

The temperature anisotropies in the CMB had a severe consequence that made the CMB photons linearly polarized. A detailed study of the sources of these polarization and the obtained spectrum can open doors to new physics and might even close for a few.

Thomson scattering, the elastic scattering of photons with electrons (here), plays a key role in the polarization of the CMB photons, and the cross section depends on polarization as

$$\frac{d\sigma_T}{d\Omega} \propto |\epsilon \cdot \epsilon'|^2$$

where ϵ (ϵ') are the incident (scattered) polarization directions. The incident light sets up oscillations of the target electron in the direction of the electric field vector $\mathbf{E}$, i.e. the polarization. The scattered radiation intensity thus peaks in the direction normal to, with polarization parallel to, the incident polarization. By the very nature of the production of this polarization, its linear nature can be realized. It can also be observed from Figure 5 that an isotropic radiation field would have changed the mechanism to give an unpolarized light. One radiation field that can produce this polarization is a *Quadrupolar radiation field* (the peaks of “hot”

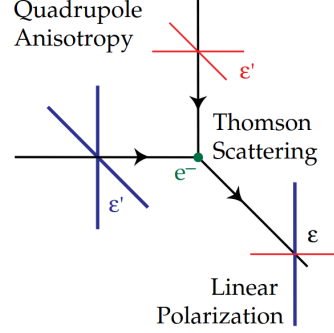


Figure 5: Production of polarization via Thomson Scattering (Image credits: [8])

and “cold” spots have an angular variation of $\pi/2$), and due to the orthogonality of spherical harmonics, these are the only ones responsible. The polarization of the photons have a major contribution from the scattering that occurred near the Surface of Last Scattering, because that was the time when the mean free path of photon increased and the cancellation of the net polarization effect due to repetitive scattering decreased, and one of the reasons why the study of the CMB Polarization is important is because it contains important information of this particular instant of time. When one tries to understand the polarization state of EM radiation, he realizes that not all polarization states are the same. Stokes Parameters is what determines the exact polarization state of an EM radiation. We will just provide a few equations related to the parameters and won't go into the details of this.

Stokes Parameters is a measure of the intensity and polarization of the light. Talking about the polarization specifically, we have three parameters, Q , U , V , where Q represents the linear polarization state (conventionally along x or y axis), U along a 45° tilted axis wrt x-y axis (a-b axis), and V is a measure of circular polarization but in this study, we set it to 0 due to reasons mentioned earlier. (Note: People reading the Stokes Parameter for the first time might want to read about why we need U when we already have Q , also see Figure 6)

$$Q = |E_x|^2 - |E_y|^2$$

$$U = |E_a|^2 - |E_b|^2$$

The net Linear Polarization state can be given by,

$$P = Q + iU$$

One problem arises when axis is rotated and Q and U transforms in such a way that it makes calculations difficult. A need for new parameters was,

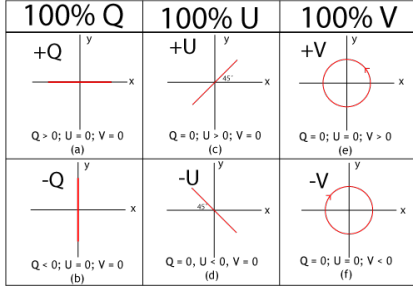


Figure 6: Visualization of the polarization part of Stokes parameters (Image credits: [Stokes parameters](#))

thus, realized that transforms like scalar. Speaking of rotations, one can perform this transformation and see that the net polarization transforms like a spin-2 field. We, therefore, use the spin weighted spherical harmonics to decompose the parameters (spin weighted because the usual spherical harmonics turned out to be insufficient for the description of polarization). The nomenclature of E and B refers to the gradient-like and curl-like nature of the electric ($\mathbf{E}$) and magnetic ($\mathbf{B}$) fields. E -modes have a gradient and zero curl, while B -modes have zero gradient and non-zero curl. Alternatively, one can see the E and B modes are the parity eigenstates of polarization. E modes have even parity (scalar) while B modes have odd parity (pseudoscalar) (see Figure 7).

$$(Q \pm iU)(\hat{n}) = \sum_l \sum_m (E_{lm} \pm iB_{lm})_{\pm 2} Y_{lm}(\hat{n})$$

The Quadrupolar variation corresponds to $l = 2$ in spherical harmonics and will have allowed multipole values of $m = 0, 1, 2$. These multipoles correspond to different kind of polarization pattern produced due to different types of perturbations; viz. scalar ($m = 0$), vector ($m = 1$), tensor ($m = 2$). Now, we will turn to one of the objectives of this article, getting a better understanding about these polarization patterns and their origin.

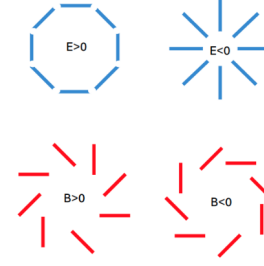


Figure 7: E and B modes of polarization (Image credits: [8])

Scalar Perturbations

These modes represent perturbations in the (energy) density of the cosmological fluid at last scattering and are the only fluctuations which can form structures through gravitational instability. Photon originating from a region having high matter-energy density will have to climb the gravitational potential well, eventually losing energy and appearing cold. Therefore, the hot regions are the regions of higher effective temperature (i.e., after taking potential into account also) and cold ones of lower, and at the scale where gravity dominates, matter (and photons) moves towards the cold region. This becomes the case of adiabatic fluctuations. We will represent the Temperature field as a Fourier expansion, giving us the intuition of the movement of the field, denoted by the wavevector k , with planewave basis. Considering only one Fourier component of this expansion, imagine an observer at the trough of the wave, i.e. at the “cold-plane” (see Figure 8). Remember that matter and photons are still coupled. The azimuthal symmetry in the problem requires that $v \parallel k$ and hence the flow is irrotational $\nabla \times v = 0$. Because hotter photons from the crests flow into the trough from the $\pm k$ directions while cold photons surround the observer in the plane, the quadrupole pattern seen in a trough has an $m = 0$ structure. Mathematically,

$$Y_2^0 \propto 3\cos^2(\theta) - 1$$

where $\theta = \hat{n} \cdot k$. Given the irrotational condition, scalar perturbations can't produce B -modes. As an interesting exercise, you can try to (qualitatively) prove that the polarization, in this case, would be either parallel or perpendicular to the direction of propagation of the wave. Nature of the modes in Figure 7 follows from this conclusion, that for a wavevector originating from the center (of the patterns) and pointing radially outwards, the E modes will be either parallel or perpendicular to it, while B mode being at an angle of 45° . It is also important

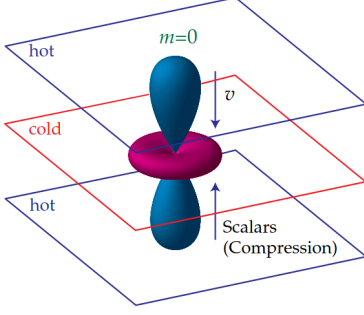


Figure 8: Polarization from Scalar Perturbations (Image credits: [6])

to note that the largest polarization signal occurs at the peaks of the temperature signal, so scalar perturbations produce a non-zero correlation between the temperature and polarization anisotropies.

Vector Perturbations

Vector perturbations represent vortical motions of the matter, where the velocity field v obeys $\nabla \cdot v = 0$ and $\nabla \times v \neq 0$. For a plane wave perturbation, the velocity field $v \perp k$ with direction reversing in crests and troughs. The radiation field at these extrema possesses a dipole pattern due to the Doppler shift from the bulk motion. Quadrupole variations vanish here but peak between velocity extrema. To see this, imagine sitting between crests and troughs. Looking up toward the trough, one sees the dipole pattern projected as a hot and cold spot across the zenith; looking down toward the crest, one sees the projected dipole reversed. The net effect is a quadrupole pattern in temperature with $m = \pm 1$,

$$Y_2^{\pm 1} \propto \sin(\theta) \cos(\theta) e^{\pm i\phi}$$

The lobes are oriented at 45° from k and v since the line of sight velocity vanishes along k and at 90° to k here. The latter follows since midway between the crests and troughs v itself is zero. The polarization produced will be largest at the nulls and at an angle $\pm 45^\circ$ to the wave direction. A key difference to the last case is that while gravity amplifies scalar perturbations, any vector perturbations will decay in an expanding universe as the inverse square of the scale factor.

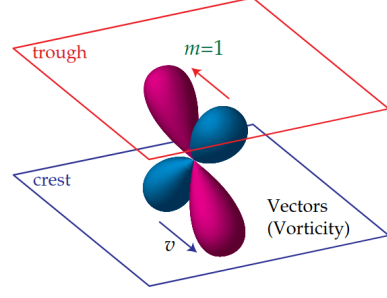


Figure 9: Polarization from Vector Perturbations (Image credits: [6])

Tensor Perturbations

Tensor fluctuations are transverse-traceless perturbations to the metric, which can be viewed as gravitational waves. A plane gravitational wave perturbation represents a quadrupolar “stretching” of space in the plane of the perturbation. Consider a circular group of test particles with a gravitational wave propagating in the y -direction normal to the xz -plane of the circle. The $+$ polarization alternately compresses and stretches the spacetime in horizontal and vertical directions causing the circle to be alternately elongated in top-to-bottom and side-to-side directions. The $\times$ polarization distorts the spacetime in a 45° alignment with respect to the $+$ polarization. The rarefaction and compression of the spacetime translates to redshift and blueshift of the photons in these directions, representing a temperature quadrupole moment with $l = 2, m = 2$,

$$Y_2^{\pm 2} \propto \sin^2(\theta) e^{\pm 2i\phi}$$

The largest polarization signal will be seen by an observer whose line of sight is parallel to the wave direction. A major thrust of current research is to search for these inflationary gravitational waves.

What lies ahead?

Any cosmological model will predict a particular power spectrum and a comparison with the actual one obtained can help us determine certain parameters or provide constraints which are crucial to test the validity of that model. Angular power spectrum depicts how a parameter changes across (a neighboring part of) the sky. For eg, if I move my detector a little bit away from a particular patch, I can use correlations to determine the change in the value of that parameter. These correlations are described as angular power spectrum. Figure 11 is the result from PLANCK (2015) and the fitting has been done according to the Λ CDM model, which shows its consistency.

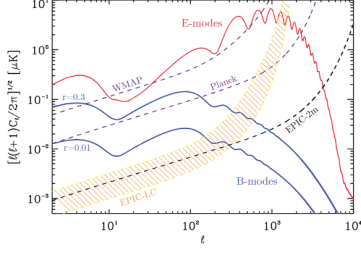
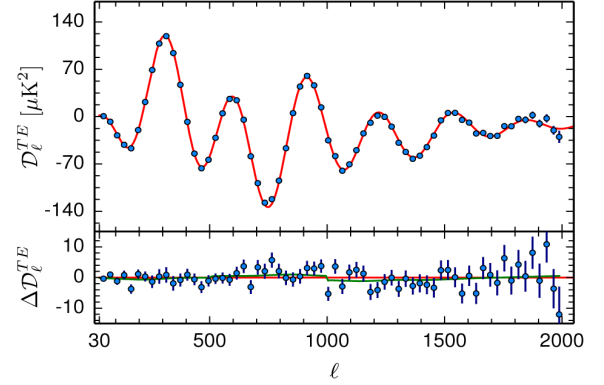
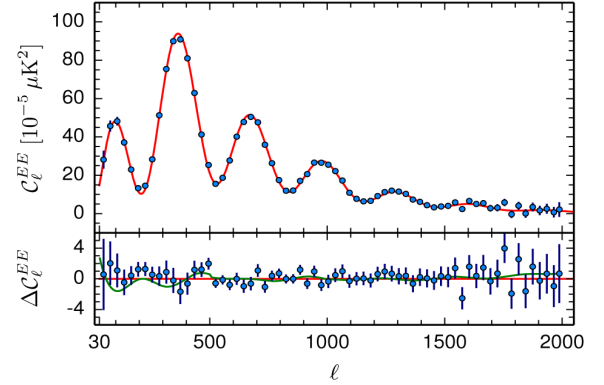


Figure 10: E and B mode power spectra for a tensor-to-scalar ratio saturating current bounds, $r = 0.3$, and for $r = 0.01$. Dotted lines are the experimental sensitivities of the instruments. (Credits: [4])

While scalar perturbations can't produce B modes, vector and tensor ones can, and due to the reasons mentioned previously, tensor ones are quite dominant. They tell us about the primordial gravitational waves, which can be further used to extract a lot of information in many areas of physics. For eg, tensor perturbations are often linked to different theories of quantum gravity and, therefore, the early universe has been transformed into a laboratory to test these theories. Lastly, it should be mentioned that B modes can also be generated via effects like Gravitational Lensing (although this is scale dependent, see Figure 10). Readers are encouraged to read more about this.



(a) Temperature-Polarization spectrum



(b) Polarization-Polarization spectrum

Figure 11: Angular Power Spectrum (Credits: [1])

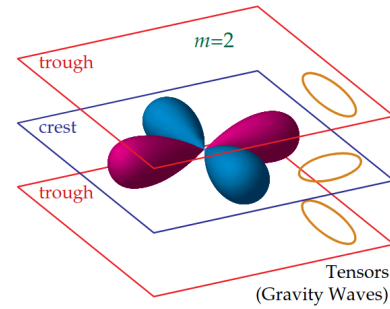


Figure 12: Polarization from Tensor Perturbations (Image credits: [6])

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